Stop cool-off: none vs 1 day vs 5 days vs signal reset

Experiment report · L/S Gamma Desk, QUANTT

2026-10-16

Question: After a stop, does waiting longer before re-entering the same side (5 trading days, or until the signal resets) beat the live 1-day cool-off on both net P&L and tail risk?

Method: Pre-registered backtest (experiment `2026-10-16-stop-cooloff`, card committed before any run). Four variants on R2 replay snapshots v1 (config_hash b040760187f0), development window 2014-06-02 to 2024-12-31, live code, one contract, at 1x and 2x spreads. Costs: $0.65 per contract, $0.005 per share, $1 minimum per order. No stop fired in the 10-year baseline, so every variant uses the same tighter stops, fixed in advance: short stop at 0.5x credit, long stop at 25% of premium. Variants: `cool0` (no cool-off), `cool1` (live rule, the reference), `cool5` (5 trading days) and `reset` (no cool-off, but no same-side entry until the signal has left that side; research only). The hold-out (2025-01-01 onward) was not used.

1 Result vs the pass bar

The card's first rule is an event floor: if `cool1` has fewer than 5 stop exits at 1x, the verdict is INCONCLUSIVE. With the tighter stops, `cool1` had **3 stops** in ten and a half years. **The verdict is INCONCLUSIVE (too few events).**

The comparisons below are shown for information only; they do not count. `cool5` and `reset` produced exactly the same trades, so they share one column.

Bar	1x	2x	Result
Event floor: `cool1` stops at 1x at least 5	3	(3)	**fails: INCONCLUSIVE**
Net P&L, full window: `cool5`/`reset` vs `cool1`	-$2,120 vs -$2,331	-$4,289 vs -$4,540	higher (info only)
Net P&L, 2014-06 to 2019-12	-$1,083 vs -$1,083	-$1,715 vs -$1,715	tie, not higher (info only)
Net P&L, 2020-01 to 2024-12	-$1,037 vs -$1,247	-$2,574 vs -$2,825	higher (info only)
ES 5% (daily), full window	-93.9 vs -98.1	-102.8 vs -107.1	less severe (info only)

Even without the event floor the hypothesis would not pass, because the first half is an exact tie. After the only stop in 2014-2019 (August 2015), the 1-day wait was already enough to avoid a same-side re-entry, so `cool1`, `cool5` and `reset` traded the same way until 2020. The whole difference comes from two stops in February and March 2020.

Full-window context at 1x (2x): baseline -$2,060 (-$4,122), ES -96.5 (-104.5); `cool0` -$2,351 (-$4,584), ES -100.7 (-110.0); `cool1` -$2,331 (-$4,540), ES -98.1 (-107.1); `cool5` and `reset` -$2,120 (-$4,289), ES -93.9 (-102.8).

2 Charts

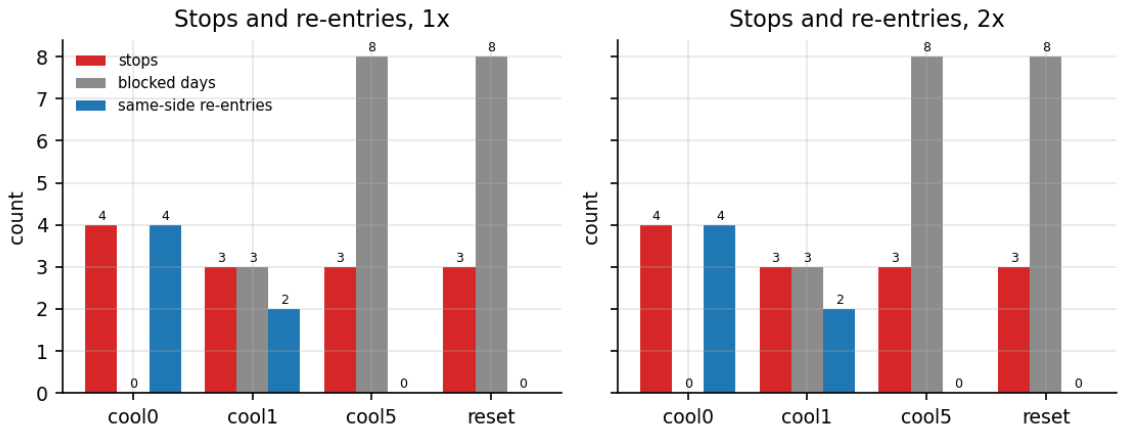


Figure 1: Stops, blocked days and same-side re-entries per variant

The tighter stops fired 3 to 4 times. With no cool-off, every stop was followed by a same-side re-entry (4 of 4). The 1-day rule blocked 3 days and still allowed 2 re-entries. Waiting 5 days or for a reset blocked 8 days and allowed none.

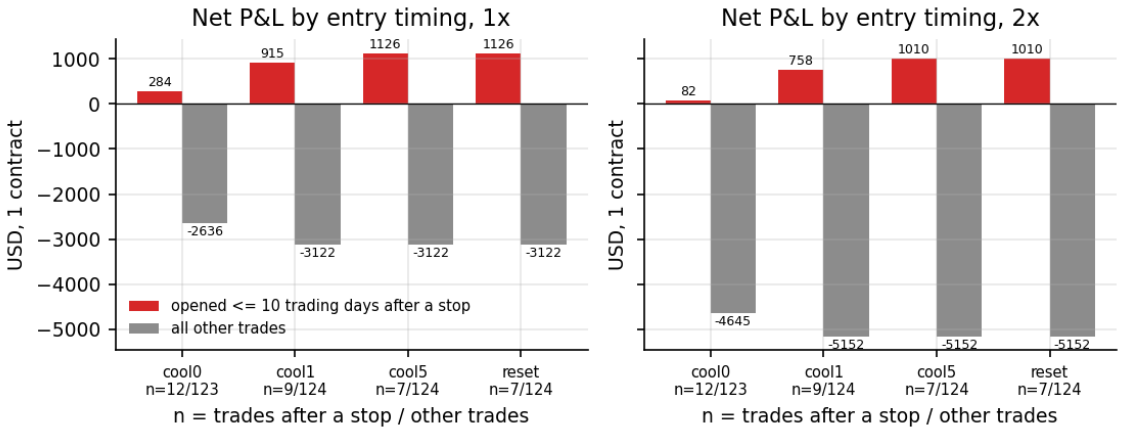


Figure 2: Net P&L of trades opened within 10 trading days after a stop vs all other trades

Trades opened soon after a stop made money in every variant, mostly in the 2020 rebound. The one exception is `cool0`'s longs (−$19 at 1x, −$129 at 2x). Waiting longer raised that group from +$915 (`cool1`) to +$1,126 at 1x. The other trades total the same for `cool1`, `cool5` and `reset` (-$3,122 at 1x), so the gap between them comes only from the post-stop window. Counts are small (7 to 12 trades).

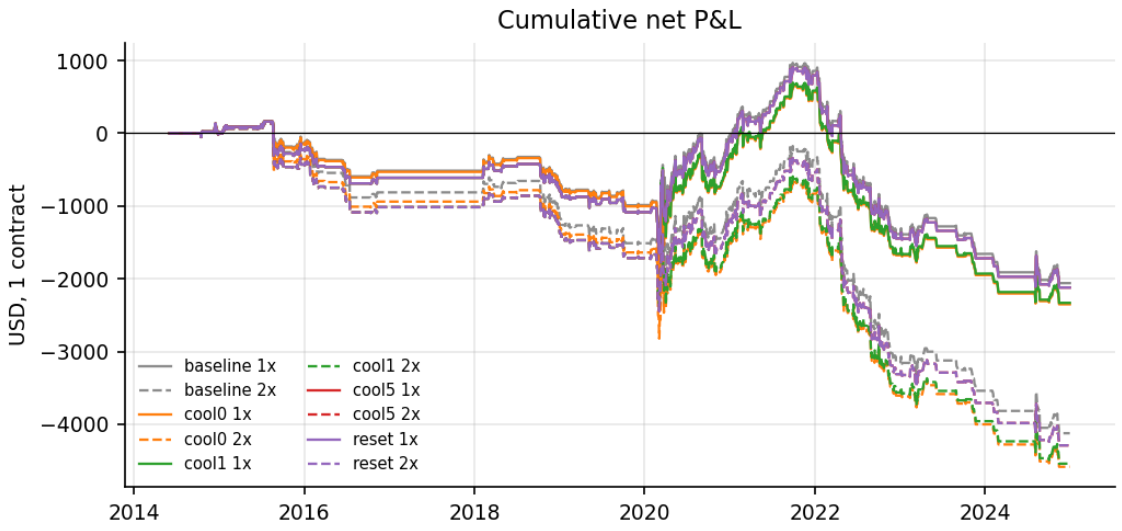


Figure 3: Cumulative net P&L, 1x solid, 2x dashed

The variants leave the baseline at the August 2015 stop (only `cool0` re-enters there). `cool1`, `cool5` and `reset` stay together until February 2020, split there, and stay apart by an almost constant amount afterwards. `cool5` is hidden under `reset` because the two are identical.

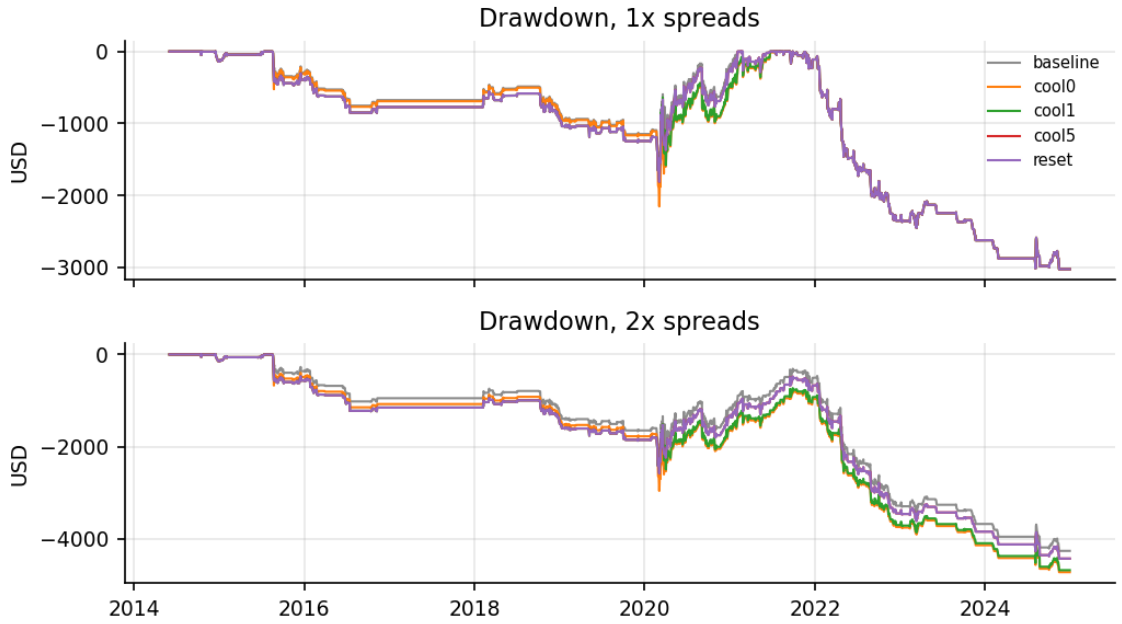


Figure 4: Drawdown at 1x and 2x

The maximum drawdown is the same for all variants at 1x ($3,027, reached at the end of 2024). The cool-off rule only changes the depth of the March 2020 dip.

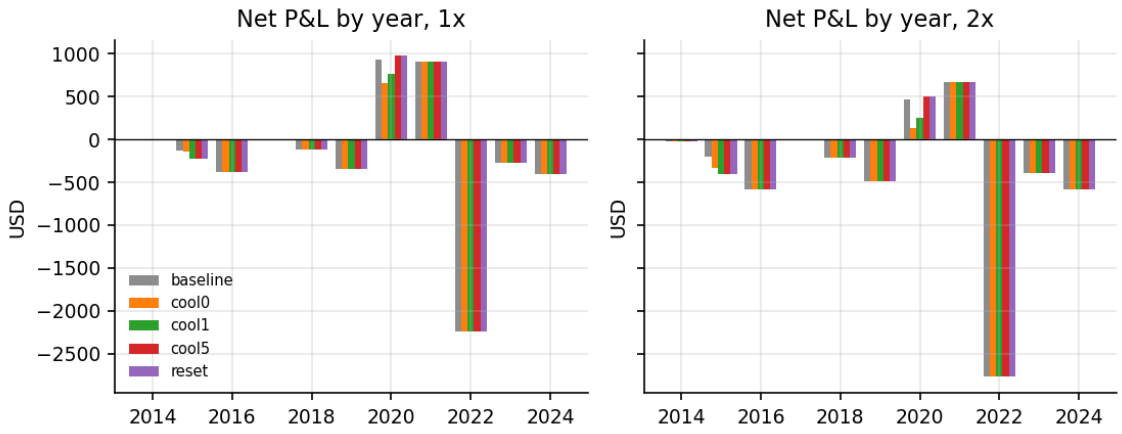


Figure 5: Net P&L by year

The cool-off rule changes only 2015 and 2020. Waiting longer was better in 2020 and made no difference in 2015.

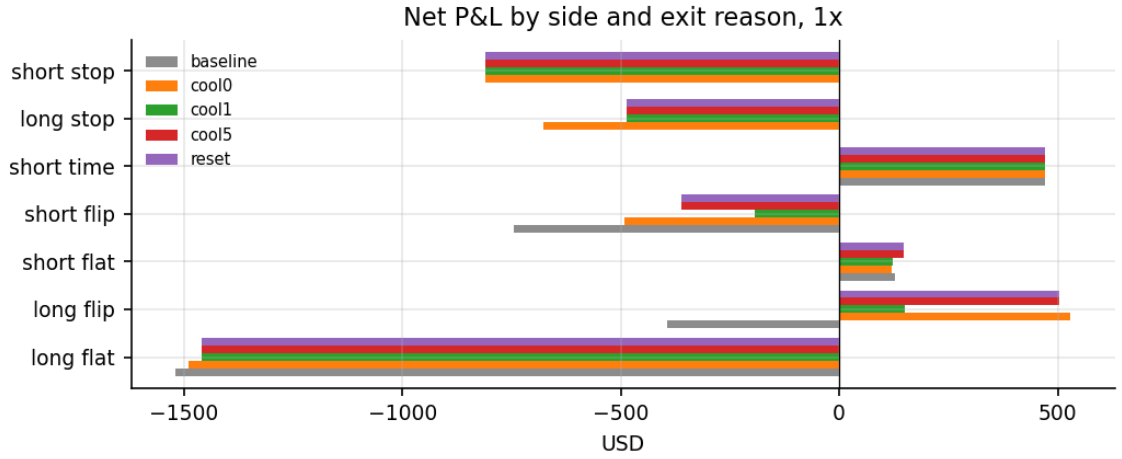


Figure 6: Net P&L by side and exit reason, 1x

The stops themselves cost $1,297 at 1x in `cool1`, `cool5` and `reset`, and $1,488 in `cool0` (4 stops). `cool0` loses more on long stops because it re-entered a long and was stopped again. The longer waits gain mainly on long flips.

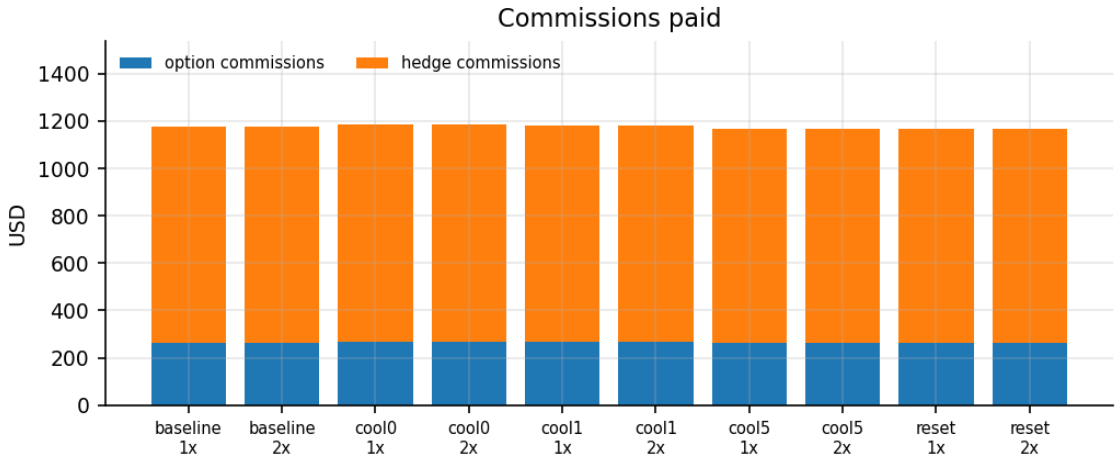


Figure 7: Commissions paid

Commissions are almost the same in every variant (about $1,170). Hedge orders and their $1 minimum dominate, so the cool-off rule has no meaningful effect on costs.

3 Analysis

The test could not answer its question, because the tighter stops fired only 3 times in ten and a half years: a short in August 2015, then a short in February 2020 and a long in March 2020. The two 2020 stops drive all the differences between the variants. The full-window direction is the one the hypothesis predicted (more P&L and a less severe ES at both spreads for the longer waits), but the mechanism is not the clean one the hypothesis assumed. The 1-day rule's two same-side re-entries in 2020 lost only $47 in total, and with no cool-off the four immediate re-entries made $169. Most of the $210 gain from waiting comes from path dependence: skipping the re-entry changed which trades came next during the 2020 rebound (mainly a better long flip). One crash and a different sequence of later trades is not evidence that waiting is better in general, and the first half shows no difference at all.

`cool5` and `reset` were identical: in each of the three episodes the two rules blocked the same 8 days and allowed the same next trade. On this data they cannot be told apart.

None of the variants beats the baseline (live stops, which never fired) at 1x: the tighter stops cost money no matter what cool-off follows them. The cool-off result is therefore conditional on stops that the desk does not run live; experiment `2026-10-16-stop-levels` tests the stop levels themselves.

4 Verdict

INCONCLUSIVE (too few events). `cool1` had 3 stop exits at 1x, below the pre-registered floor of 5. The live rule (`[exit] stop_cooloff_days = 1`) stays. The 2020 episode points towards a longer wait, but it rests on one crash and on path effects rather than on losing re-entries, and even without the floor the first-half tie would have failed the bar. A useful follow-up needs more stop events: more symbols (the R2 bucket also has QQQ and IWM options), or a stop level that fires more often. Any such test would need its own card.

5 Caveats

- **In-sample idea:** the need for this test, and the tighter stops, came from the baseline on the same window. Even a pass would have needed one hold-out run.

- **Tiny sample:** 3 stops, 2 of them in one crash, and 7 to 12 trades after a stop. The comparison rests on a handful of trades.
- **Conditional on tighter stops:** the live stops never fired, so the cool-off rule has had no effect on the live configuration in ten years.
- **One contract:** the $1 minimum per order is large relative to the position size. It dominates hedge costs, the same in every variant.
- **Spread model:** half-spreads are fitted on about two weeks of calm 2026 live quotes. They are probably too narrow in crashes, which is exactly when the stops fire.
- **Missing data:** month-end R2 days are missing; 3.3% of scheduled times were skipped.
- **Weekdays:** cool-off days are counted as weekdays, as in the live rule, so an exchange holiday counts as a day.

6 Trials

This experiment registered 8 trials (4 variants at 1x and 2x), run once on commit 8d1f0fe, with the pass bar committed before the run. Adding the baseline's 2 trials gives 10. The baseline regression check that reproduced trial 1 was not registered. `cool1` has exactly the same configuration as the `tight` variant of `2026-10-16-stop-levels`, and its results reproduce that experiment's numbers exactly (-$2,330.67 and ES -98.06 at 1x). It is a duplicate trial, accepted in advance by the PM. Some registry rows are marked "dirty" because other experiments' uncommitted research files were in the working tree during the run; this experiment's script, the configs and `src/` were unchanged from 8d1f0fe.